

DO NOT WRITE ON TEST BOOKLET



**COMPUTER PROGRAMMING
CONCEPTS
(390)**

—OPEN EVENT—

STATE – 2020

TOTAL POINTS

_____ ***(100 points)***

Failure to adhere to any of the following rules will result in disqualification:

- 1. Contestant must hand in this test booklet and all printouts. Failure to do so will result in disqualification.**
- 2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests or facsimile (handwritten, photocopied, or keyed) are allowed in the testing area.**
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No more than sixty (60) minutes testing time

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Choose the *best* answer to the following questions. Mark answers clearly if using a scantron form. Mark A if the statement is true. Mark B if the statement is false.

1. Which of the following would be a valid name for a memory location in most languages?
 - a. my.name
 - b. a1B2c3D4
 - c. me&You
 - d. 3oreos
2. C++, Visual Basic, C#, Python and Java are all examples of what kind of language?
 - a. Assembly
 - b. Machine
 - c. Mnemonic
 - d. Programming
3. What is the binary equivalent of the decimal number 7?
 - a. 1001
 - b. 0011
 - c. 0111
 - d. 1010
4. Which of the following is a scripting language?
 - a. Perl
 - b. HTML
 - c. PHP
 - d. All of the above
5. What type of structure is created when a pretest loop is inside another pretest loop?
 - a. integrated selection
 - b. nested repetition
 - c. iterative repetition
 - d. optional selection
6. What does the oval symbol in a flowchart represent?
 - a. branching
 - b. processing
 - c. input/output
 - d. start/stop
7. Which development component combines the object file with other machine code necessary for your C++ program to run correctly?
 - a. IDE
 - b. linker
 - c. object code
 - d. source code

8. Which programming language is an interpreted language most commonly used for client-side Web scripting, such as animating page elements and validating input on HTML forms?
 - a. Prolog
 - b. Python
 - c. Java
 - d. Javascript
9. Which of the following tells the computer to pass the variable's address rather than a copy of its contents?
 - a. return statement
 - b. address-of operator
 - c. value-returning function
 - d. referral operator
10. A low-level language has a low level of _____ because it includes commands specific to a particular CPU or microprocessor family.
 - a. flow control
 - b. syntax
 - c. iteration
 - d. abstraction
11. _____ perform differently based on their operands.
 - a. Boolean operators
 - b. Conditional expressions
 - c. Overloaded operators
 - d. Relational operators
12. Which of the following are logical operators in most languages?
 - a. != and ==
 - b. >, <, >=, <=
 - c. ==, =, and !=
 - d. && and ||
13. In order to write a recursive solution _____.
 - a. you must create an infinite loop
 - b. the simplest case should be determined
 - c. write the iterative solution first
 - d. the most complicated approach must be identified
14. A common problem associated with counter-controlled loops is not executing the loop for the last value. This type of problem is labeled a(n) _____ error.
 - a. counter-controlled
 - b. boundaries
 - c. last-value
 - d. off-by-one

15. Pseudocode is standardized, each programmer writes pseudocode the same.

- a. True
- b. False

16. Which symbol is used to represent the address-of operator?

- a. @
- b. &
- c. #
- d. \$

17. In the following code, what is the scope and lifetime of the num1 variable?

```
int num2 = 100
main( )
{
    for (int num1=1; num1 < num2; num1+=1)
    {
        .....more code here
    }
}
```

- a. global, until the for loop ends
- b. local, until the program ends
- c. local, until the for loop ends
- d. global, until the program ends

18. The code below is considered a(n) _____.

```
if (examScore > 89)
    grade = 'A'
```

- a. iterative statement
- b. one-way if statement
- c. two-way if statement
- d. simple sequential statement

19. Using the code below, when does grade get assigned a value of 'A'?

```
if (examScore > 89);
    grade = 'A'
```

- a. Whenever the score is greater than 89
- b. When the score is less than or equal to 89
- c. Never, a syntax error is generated
- d. Every time the program is run

20. Using the code below, when does “Excellent” get displayed?

```
if (examScore > 89)
    grade = 'A';
    println("Excellent");
```

- a. Whenever the score is greater than 89
- b. When the score is less than or equal to 89
- c. Never, a syntax error is generated
- d. Every time the program is run

21. What happens when *val* is equal to *largest* in the program segment below?

```
if (val < largest)
    result = val
else
    result = largest
```

- a. Result is assigned *val*
- b. Result is assigned *largest*
- c. Nothing, neither assignment statement is executed
- d. An error is reported

22. Which of the following statements is NOT true regarding switch statements?

- a. The expression is sometimes called the selector.
- b. Default is only executed when there is no case match.
- c. A variable cannot be used as the case label.
- d. Default is required.

23. What is the first value printed in the program segment below?

```
int counter = 0
while (counter < 100)
{   println(counter)
    counter++
}
```

- a. 0
- b. 1
- c. 99
- d. 100

24. A method that calls itself repeatedly until it arrives at the solution is a(n) _____ method.

- a. static
- b. iterative
- c. recursive
- d. instance

25. What is the last value printed in the program segment below?

```
int counter = 0
while (counter < 100)
{   println(counter)
    counter++
}
```

- a. 0
- b. 1
- c. 99
- d. 100

26. How many times would the loop body executed if changed to the following?

```
int counter = 0
while (counter < 100)
{   println(counter)
    counter+=5
}
```

- a. 19
- b. 20
- c. 99
- d. 100

27. What happens if the *break* is omitted in the code below?

```
switch (phoneDigit)
{
    case 1:    num = 1;
               break;
    case 2:    num = 2;
               break;
    case 3:    num = 3;
               break;
    case 4:    num = 4;
               break;
    default:   num = 0;
               break;
}
```

- a. A syntax error is generated.
- b. num is never assigned a value
- c. num is always assigned 0.
- d. num is assigned 1.

28. What value is stored in num if the *phoneDigit* has a value of 'Q' in the code below?

```
switch (phoneDigit)
{
    case 1:    num = 1;
               break;
    case 2:    num = 2;
               break;
    case 3:    num = 3;
               break;
    case 4:    num = 4;
               break;
    default:   num = 0;
               break;
}
```

- a. 0
- b. 1
- c. Q
- d. Not enough information given.

29. What is TRUE about the variable z as defined in the following code segment?

```
for(int z = 0; z < 10; z++)
{
    println("counter" + z)
}
```

- a. z is out of scope when the loop terminates.
- b. z creates an error because it is changed in the update portion
- c. z would have a value of 10 if it were printed outside of the loop.
- d. z can be displayed prior to the loop segment

30. How many times will the loop body be executed in the following code segment?

```
for(int z = 0; z < 10; z++)
{
    println("counter" + z)
}
```

- a. 4
- b. 5
- c. 6
- d. 10

31. Which of the following statements regarding curly braces is NOT correct as they relate to their use with an *if* statement?

- a. They can be added to increase readability.
- b. They can be used with the true statements and omitted with the false statements.
- c. They are required when there is more than one statement following the *if* or *else*.
- d. They are used to block three or more statements in an *if* or *else*.

32. What is a factor that remains the same throughout a program?

- a. object
- b. constant
- c. variable
- d. character

33. What is a graphical representation of the way a computer should progress from one instruction to the next when it performs a task?

- a. pseudocode
- b. paradigm
- c. VDE
- d. flowchart

34. What is the output of the following algorithm if AMOUNT is equal to \$250?

```

if AMOUNT is greater than $500
    print "That's too much."
else if AMOUNT is greater than $200
    print "Still too much."
else if AMOUNT is greater than $100
    print "That's about right."
else
    print "Amount is too low."
End if
    
```

- a. That's too much.
- b. Still too much.
- c. That's about right.
- d. Amount is too low.

35. Identify the type of structure used in the following algorithm.

```

if      time-of-day is greater than noon
        set afternoon access permission
else    set regular access permission
    
```

- a. selection
- b. sequence
- c. iterative
- d. infinite

36. What is the output of the following code segment?

```
int num = 10
while (num > 10
    num = num - 2
print num
```

- a. 10
- b. 8
- c. 0
- d. None of these

37. Which of the following translates all of a program's high level instructions into machine language before executing the program?

- a. interpreter
- b. compiler
- c. translator
- d. executor

38. Comparison operators evaluate to what type of value?

- a. boolean
- b. character
- c. integer
- d. real

39. Which numerical base includes the use of Roman letters to express numerical values?

- a. Binary
- b. Decimal
- c. Hexidecimal
- d. Octal

40. Dividing a problem or task into parts is an example of _____ .

- a. Abstraction
- b. Compilation
- c. Encapsulation
- d. Generality

41. What stores the information passed to the procedure or method when the procedure or method is invoked or called?

- a. Constraints
- b. Events
- c. Parameters
- d. Variables

42. What is the output of the code segment below?

```
if ( 6 is greater than 8)
    print ("**")
    print ("****")
else if ( 9 equals 4)
    print ("****")
else
    print ("*")
```

- a. ****
- b. ***
- c. **
- d. *

43. What does the process of locating and correcting the errors in a program called?

- a. defragging
- b. debugging
- c. deflating
- d. extrapolating

44. What is the value of p after the following pseudocode is executed?

```
a = 2      b = 2      c = 4
if ( a > b )
    if ( a > c )
        p = 1
    else
        p = 2          end if
else
    if ( c != 0 )
        p = 3
    else
        p = 4          end if
end if
```

- a. 1
- b. 2
- c. 3
- d. 4

45. What is the term used for the assignment of a beginning value to a memory location?

- a. implicit type conversion
- b. initializing
- c. explicit type conversion
- d. revision

46. Which type of structure do ALL programs use?

- a. repetition
- b. selection
- c. sequential
- d. variable

47. The entry inside the parentheses of the following statement is called the _____.

```
if (examScore is less than 50)
    display message "Do better next time."
```

- a. selection expression
- b. statement body
- c. truth value
- d. conditional expression

48. The switch statement case value can evaluate to all of the following EXCEPT _____.

- a. char
- b. double
- c. int
- d. string

49. Which of the following is NOT an example of a loop?

- a. do while
- b. for
- c. if else
- d. while

50. Comments that use two forward slashes are called _____.

- a. block
- b. inline
- c. multiline
- d. XML